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Java SE 27 & JDK 27

Interface WildcardType

All Superinterfaces:

Type

```
public interface WildcardType
extends Type
```

WildcardType represents a wildcard type expression, such as `?`, `? extends Number`, or `? super Integer`.

See *Java Language Specification*:

4.5.1 Type Arguments of Parameterized Types

Since:

1.5

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
`Type[]`	**`getLowerBounds()`**	Returns an array of Type objects representing the lower bound(s) of this type variable.
`Type[]`	**`getUpperBounds()`**	Returns an array of Type objects representing the upper bound(s) of this type variable.

Methods declared in interface Type

`getTypeName`

Method Details

getUpperBounds

```
Type[] getUpperBounds()
```

Returns an array of Type objects representing the upper bound(s) of this type variable. If no upper bound is explicitly declared, the upper bound is `Object`.

For each upper bound B :

- if B is a parameterized type or a type variable, it is created, (see `ParameterizedType` for the details of the creation process for parameterized types).
- Otherwise, B is resolved.

API Note:

While to date a wildcard may have at most one upper bound, callers of this method should be written to accommodate multiple bounds.

Returns:

an array of Types representing the upper bound(s) of this type variable

Throws:

`TypeNotPresentException` - if any of the bounds refers to a non-existent type declaration

`MalformedParameterizedTypeException` - if any of the bounds refer to a parameterized type that cannot be instantiated for any reason

getLowerBounds

`Type[] getLowerBounds()`

Returns an array of `Type` objects representing the lower bound(s) of this type variable. If no lower bound is explicitly declared, the lower bound is the type of `null`. In this case, a zero length array is returned.

For each lower bound B :

- if B is a parameterized type or a type variable, it is created, (see `ParameterizedType` for the details of the creation process for parameterized types).
- Otherwise, B is resolved.

API Note:

While to date a wildcard may have at most one lower bound, callers of this method should be written to accommodate multiple bounds.

Returns:

an array of Types representing the lower bound(s) of this type variable

Throws:

`TypeNotPresentException` - if any of the bounds refers to a non-existent type declaration

`MalformedParameterizedTypeException` - if any of the bounds refer to a parameterized type that cannot be instantiated for any reason

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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